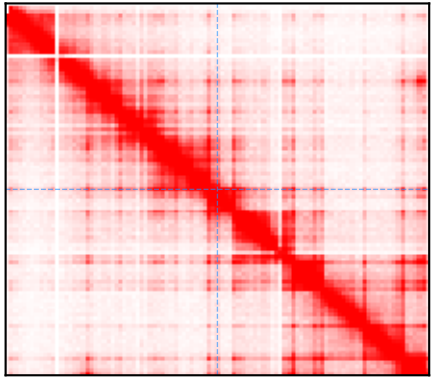
Anchor: chr1:1,019,276–1,019,776 anc_REF: 9,261 anc_ALT: 8,420

AlphaGenome pred: -0.0204 | empiric ChIP-seq: 1.17 | Hi-C: +0.14

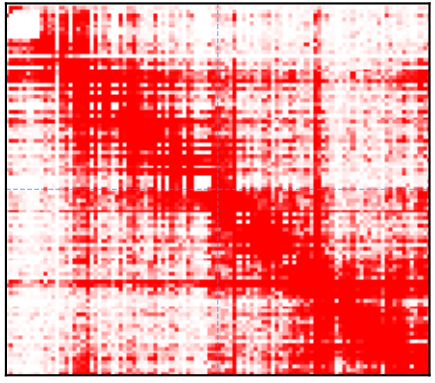
Unphased Σ=48,724

View: 965,500 – 1,065,500
Loop: 1,010,000 – 1,021,000

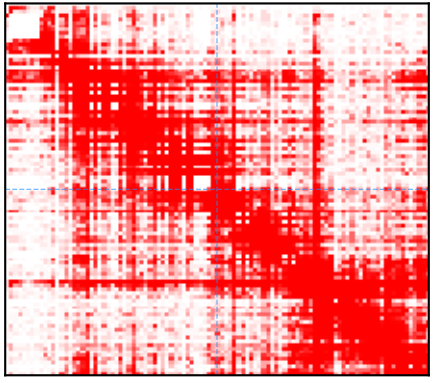
Loop 1
chr1:1,013,387-1,013,486
× chr1:1,010,000-1,021,000
dist: 2 kb
R/A: 0.96× log2FC: -0.06



Ref (hap1) Σ=1,539



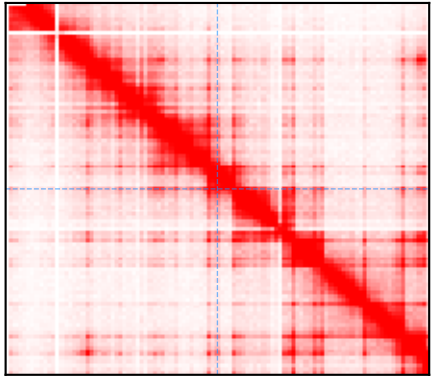
Alt (hap2) Σ=1,607



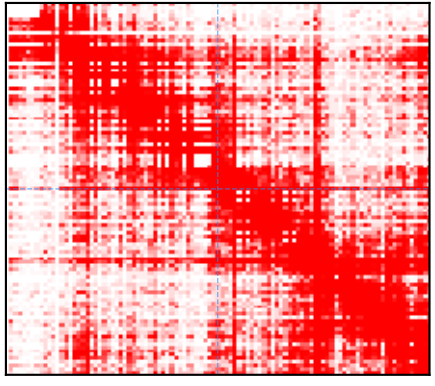
Unphased Σ=29,231

View: 965,500 – 1,065,500
Loop: 1,010,000 – 1,021,000

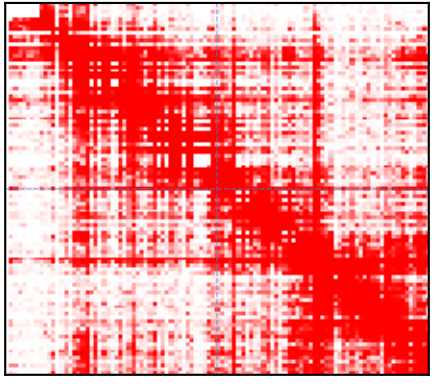
Loop 2
chr1:1,019,275-1,019,776
× chr1:1,010,000-1,021,000
dist: 4 kb
R/A: 1.13× log2FC: +0.18



Ref (hap1) Σ=7,722



Alt (hap2) Σ=6,813



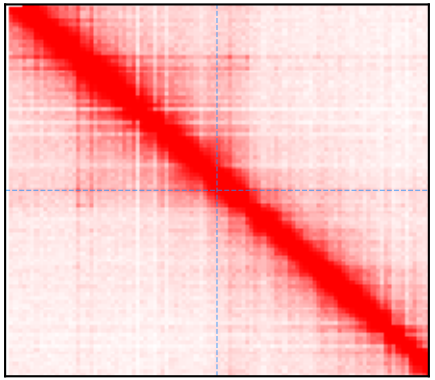
Anchor: chr10:29,449,156–29,449,250 anc_REF: 15,537 anc_ALT: 13,197

AlphaGenome pred: 0.00333 | empiric ChIP-seq: 4 | Hi-C: +0.24

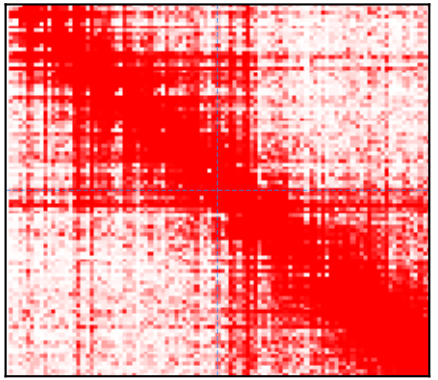
Unphased Σ=34,153

View: 29,392,500 – 29,492,500
Loop: 29,435,000 – 29,450,000

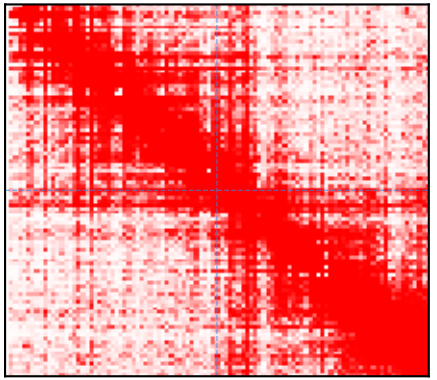
Loop 1
chr10:29,445,630-29,445,679
× chr10:29,435,000-29,450,000
dist: 3 kb
R/A: 1.24× log2FC: +0.31



Ref (hap1) Σ=7,798



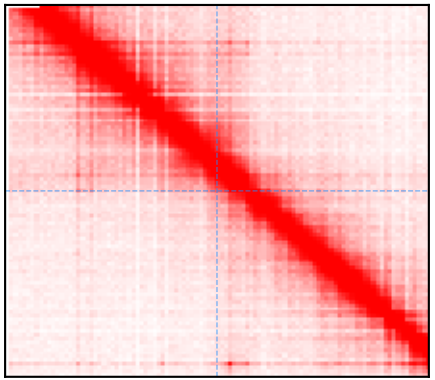
Alt (hap2) Σ=6,305



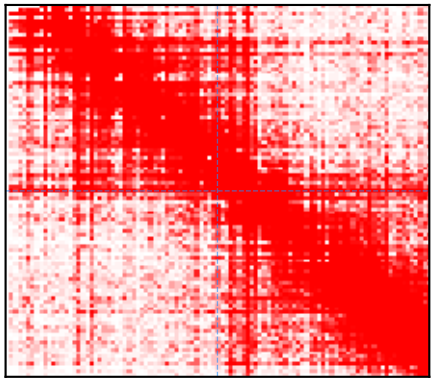
Unphased Σ=17,365

View: 29,392,500 – 29,492,500
Loop: 29,435,000 – 29,450,000

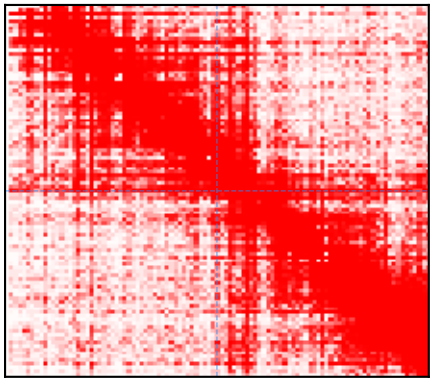
Loop 2
chr10:29,449,155-29,449,250
× chr10:29,435,000-29,450,000
dist: 7 kb
R/A: 1.12× log2FC: +0.17



Ref (hap1) Σ=7,739



Alt (hap2) Σ=6,892



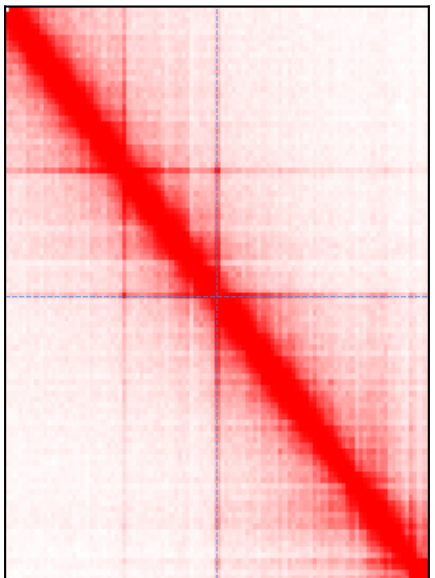
Anchor: chr12:49,966,932–49,967,728 anc_REF: 4,783 anc_ALT: 4,920

AlphaGenome pred: 0.00164 | empiric ChIP-seq: -2.25 | Hi-C: -0.04

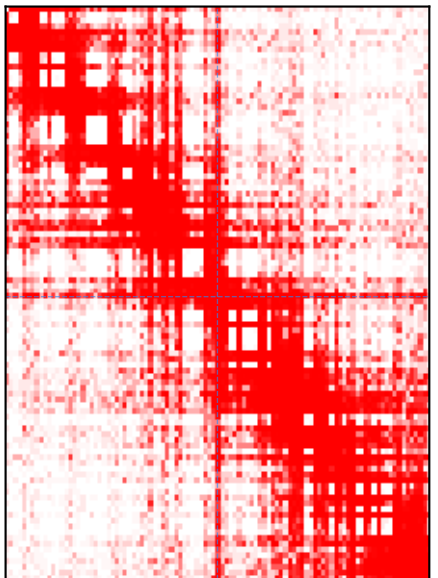
Unphased Σ=28,842

View: 49,917,000 – 50,017,000
Loop: 49,966,000 – 49,968,000

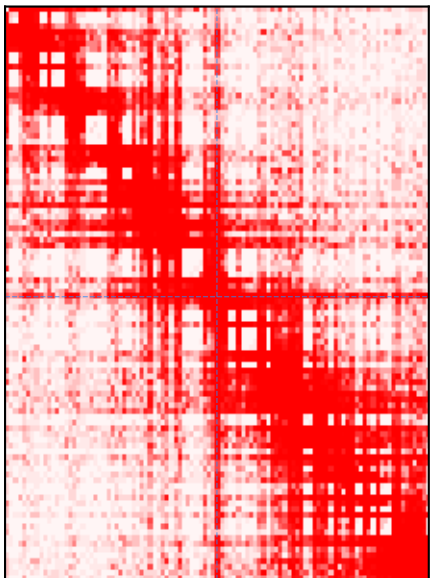
Loop 1
chr12:49,966,931-49,967,728
× chr12:49,966,000-49,968,000
dist: 0 kb
R/A: 0.97× log2FC: -0.04



Ref (hap1) Σ=4,783



Alt (hap2) Σ=4,920



Anchor: chr14:22,556,999–22,557,483 anc_REF: 28,027 anc_ALT: 26,217

AlphaGenome pred: -0.0172 | empiric ChIP-seq: 1.4 | Hi-C: +0.10

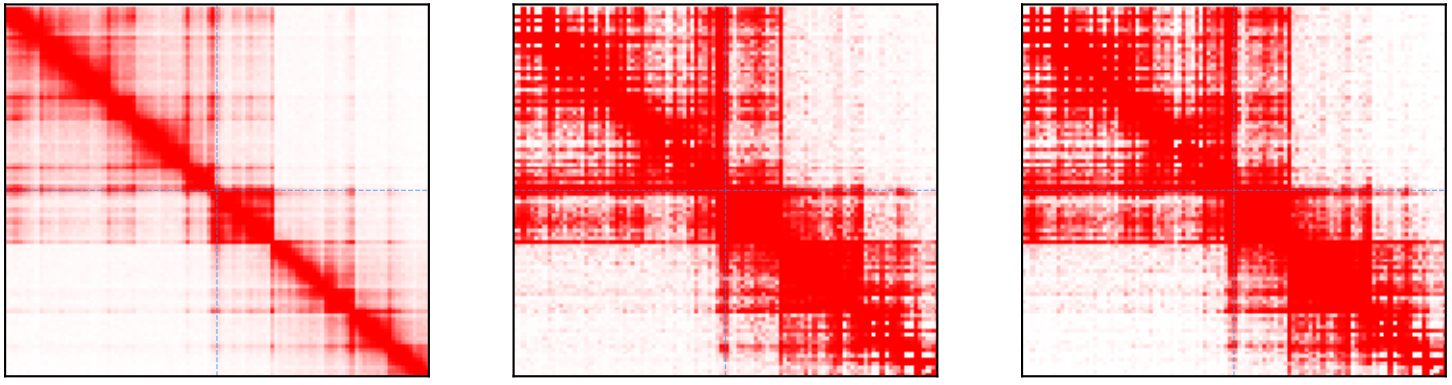
Unphased Σ=29,492

View: 22,507,000 – 22,607,000
Loop: 22,556,000 – 22,558,000

Ref (hap1) Σ=11,849

Alt (hap2) Σ=11,028

Loop 1
chr14:22,556,264-22,556,653
× chr14:22,556,000-22,558,000
dist: 1 kb
R/A: 1.07× log2FC: +0.10



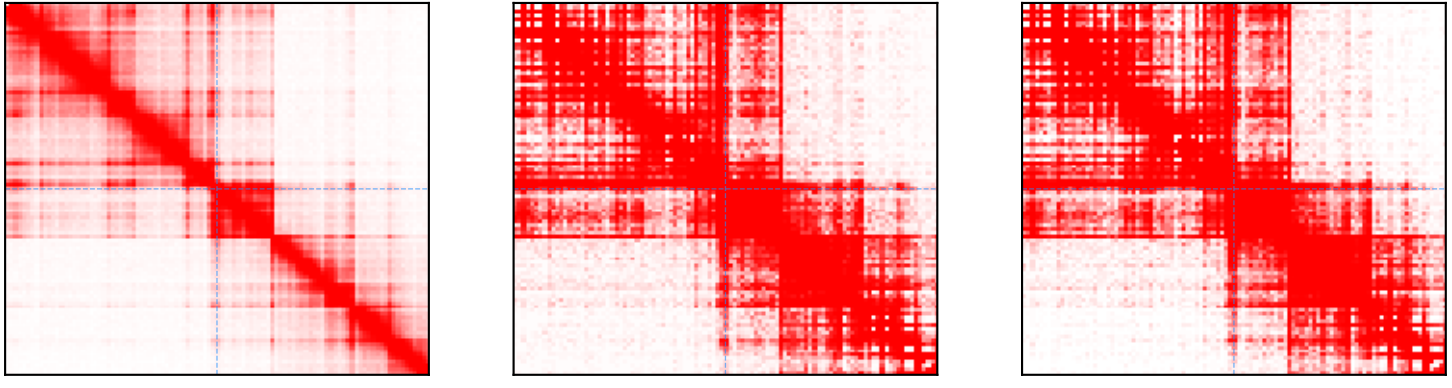
Unphased Σ=41,764

View: 22,507,000 – 22,607,000
Loop: 22,556,000 – 22,558,000

Ref (hap1) Σ=16,178

Alt (hap2) Σ=15,189

Loop 2
chr14:22,556,998-22,557,483
× chr14:22,556,000-22,558,000
dist: 0 kb
R/A: 1.07× log2FC: +0.09



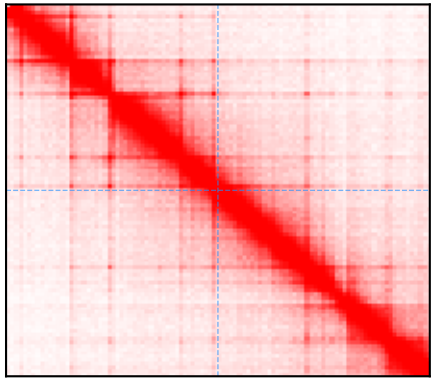
Anchor: chr14:106,564,201–106,564,275 anc_REF: 3,591 anc_ALT: 4,420

AlphaGenome pred: 0.007 | empiric ChIP-seq: -4.17 | Hi-C: -0.30

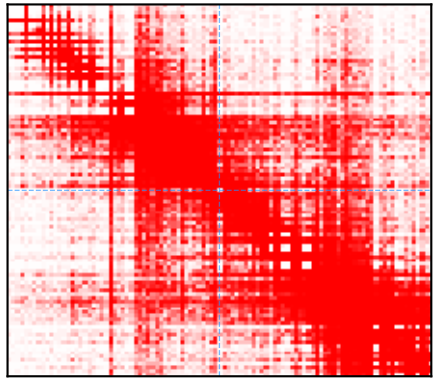
Unphased Σ=18,536

View: 106,515,000 – 106,615,000
Loop: 106,564,000 – 106,566,000

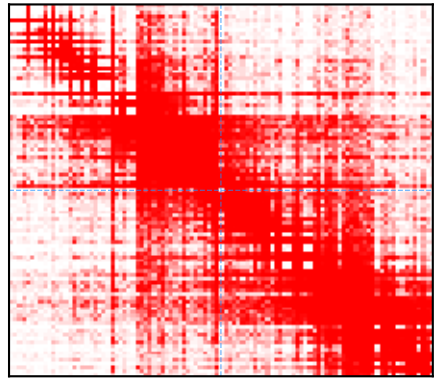
Loop 1
chr14:106,565,460-106,565,670
× chr14:106,564,000-106,566,000
dist: 1 kb
R/A: 0.80× log2FC: -0.33



Ref (hap1) Σ=1,475



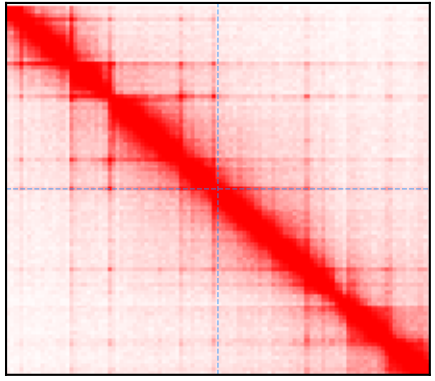
Alt (hap2) Σ=1,850



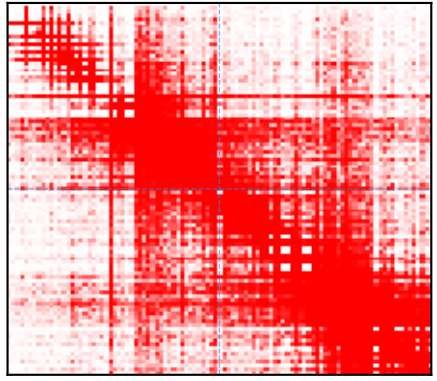
Unphased Σ=12,735

View: 106,515,000 – 106,615,000
Loop: 106,564,000 – 106,566,000

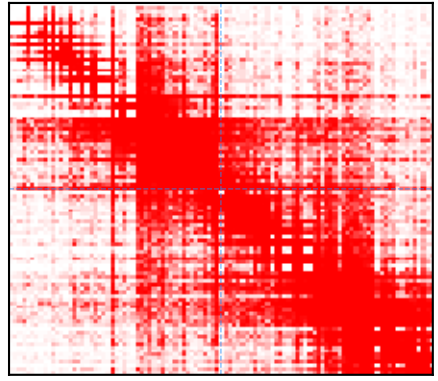
Loop 2
chr14:106,564,200-106,564,275
× chr14:106,564,000-106,566,000
dist: 1 kb
R/A: 0.82× log2FC: -0.28



Ref (hap1) Σ=2,116



Alt (hap2) Σ=2,570



Anchor: chr16:10,256,830–10,257,260 anc_REF: 20,251 anc_ALT: 18,667

AlphaGenome pred: -0.023 | empiric ChIP-seq: 1.16 | Hi-C: +0.12

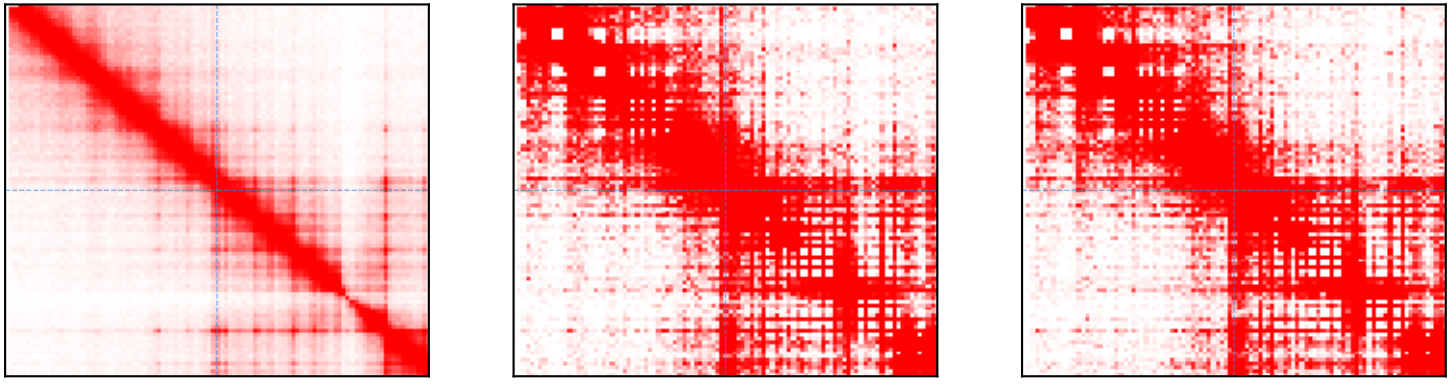
Unphased Σ=26,502

View: 10,207,500 – 10,307,500
Loop: 10,255,000 – 10,260,000

Ref (hap1) Σ=9,856

Alt (hap2) Σ=9,240

Loop 1
chr16:10,259,208-10,259,227
× chr16:10,255,000-10,260,000
dist: 2 kb
R/A: 1.07× log2FC: +0.09



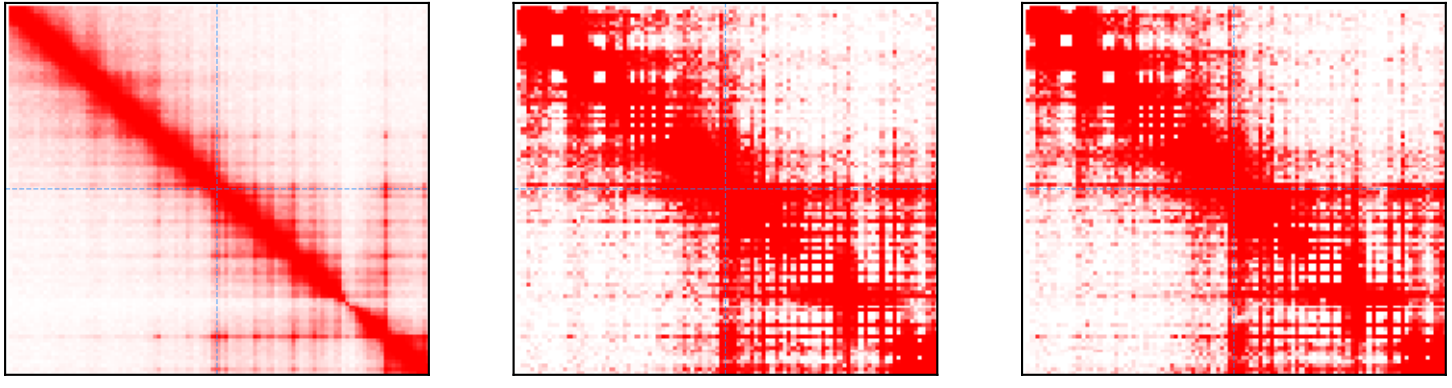
Unphased Σ=38,588

View: 10,207,500 – 10,307,500
Loop: 10,255,000 – 10,260,000

Ref (hap1) Σ=10,395

Alt (hap2) Σ=9,427

Loop 2
chr16:10,256,829-10,257,260
× chr16:10,255,000-10,260,000
dist: 0 kb
R/A: 1.10× log2FC: +0.14



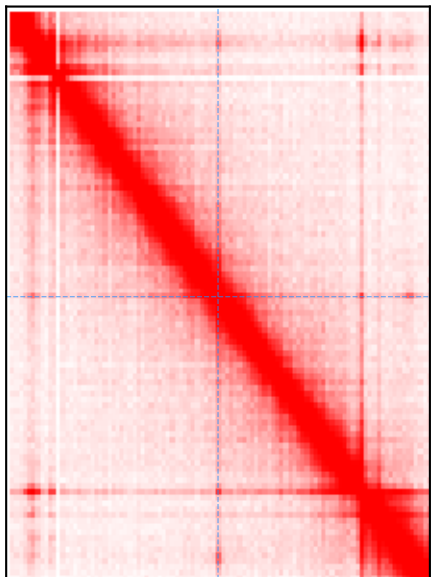
Anchor: chr17:3,602,688–3,602,760 anc_REF: 4,392 anc_ALT: 4,507

AlphaGenome pred: -0.0574 | empiric ChIP-seq: 1.32 | Hi-C: -0.04

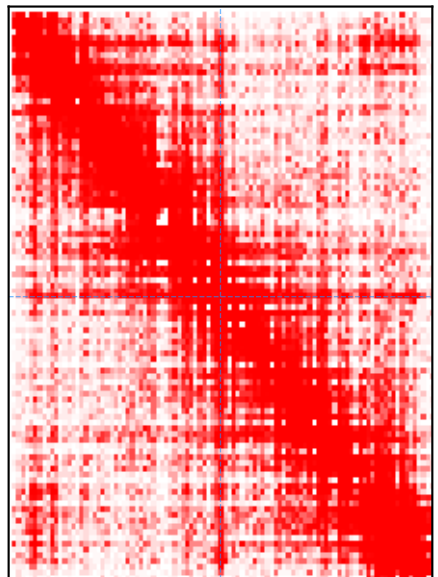
Unphased Σ=16,085

View: 3,552,500 – 3,652,500
Loop: 3,601,500 – 3,603,500

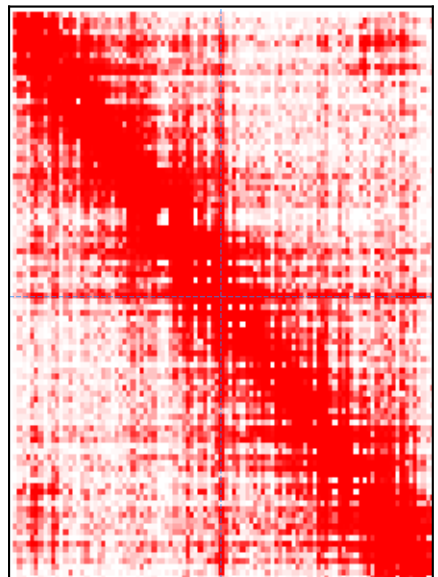
Loop 1
chr17:3,602,687-3,602,760
× chr17:3,601,500-3,603,500
dist: 0 kb
R/A: 0.97× log2FC: -0.04



Ref (hap1) Σ=4,392



Alt (hap2) Σ=4,507



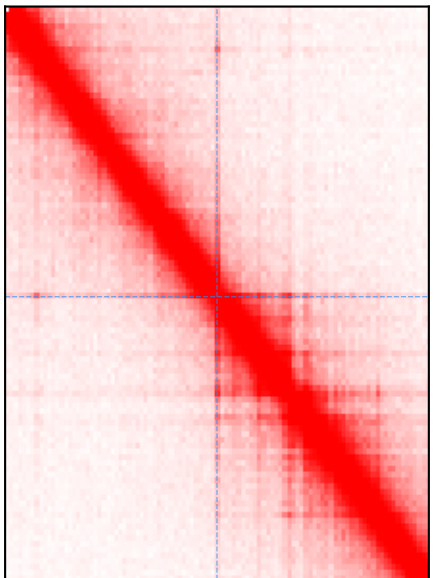
Anchor: chr17:34,361,633–34,361,717 anc_REF: 3,388 anc_ALT: 4,196

AlphaGenome pred: -0.0583 | empiric ChIP-seq: -4.91 | Hi-C: -0.31

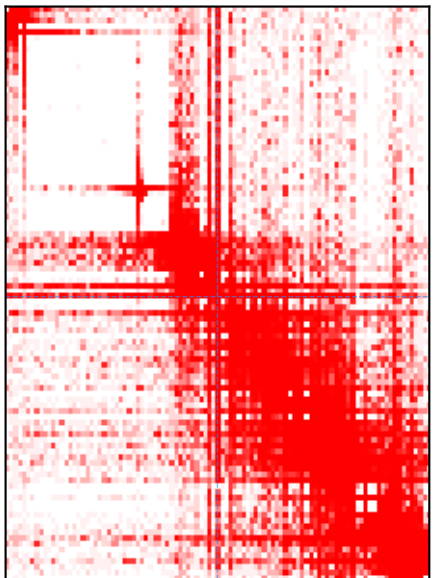
Unphased Σ=18,558

View: 34,311,000 – 34,411,000
Loop: 34,360,000 – 34,362,000

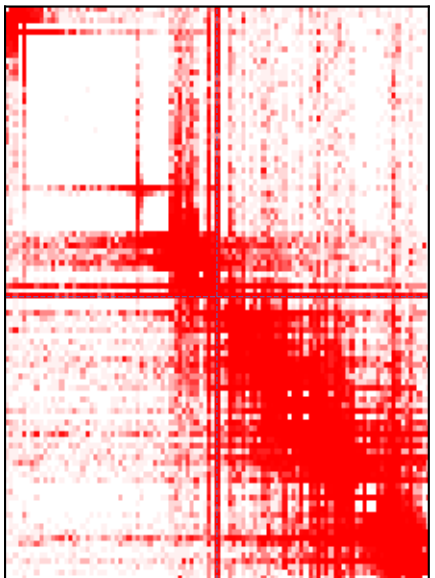
Loop 1
chr17:34,361,632-34,361,717
× chr17:34,360,000-34,362,000
dist: 1 kb
R/A: 0.81× log2FC: -0.31



Ref (hap1) Σ=3,388



Alt (hap2) Σ=4,196



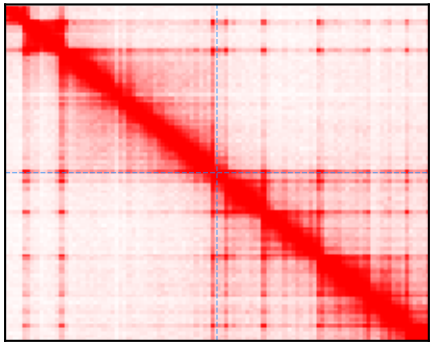
Anchor: chr17:82,494,711–82,494,907 anc_REF: 20,486 anc_ALT: 18,324

AlphaGenome pred: 0 | empiric ChIP-seq: 2.91 | Hi-C: +0.16

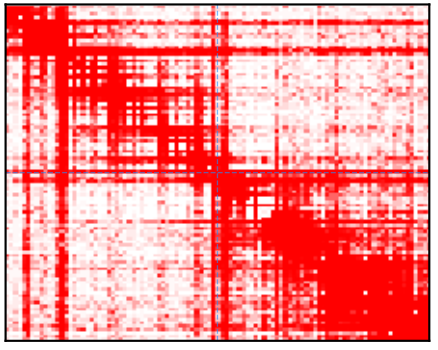
Unphased Σ=47,797

View: 82,445,000 – 82,545,000
Loop: 82,490,000 – 82,500,000

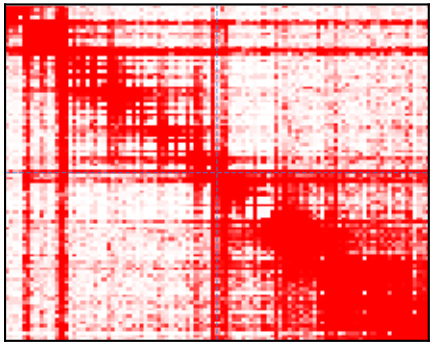
Loop 1
chr17:82,494,710-82,494,907
× chr17:82,490,000-82,500,000
dist: 0 kb
R/A: 1.11× log2FC: +0.15



Ref (hap1) Σ=5,788



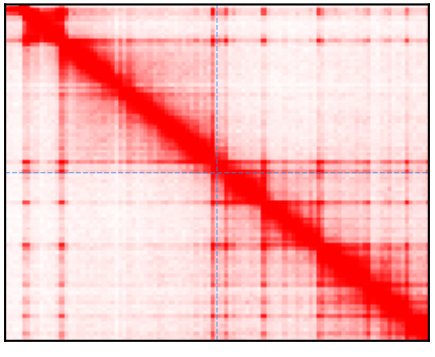
Alt (hap2) Σ=5,232



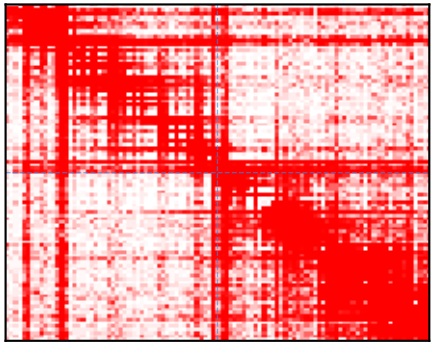
Unphased Σ=43,212

View: 82,445,000 – 82,545,000
Loop: 82,490,000 – 82,500,000

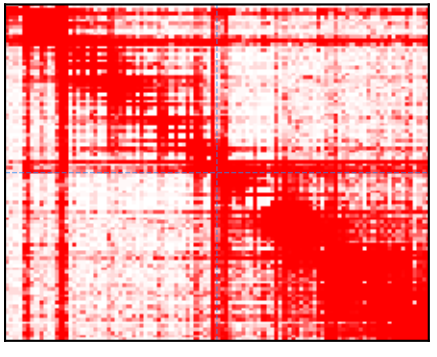
Loop 2
chr17:82,497,270-82,497,289
× chr17:82,490,000-82,500,000
dist: 2 kb
R/A: 1.15× log2FC: +0.20



Ref (hap1) Σ=11,361



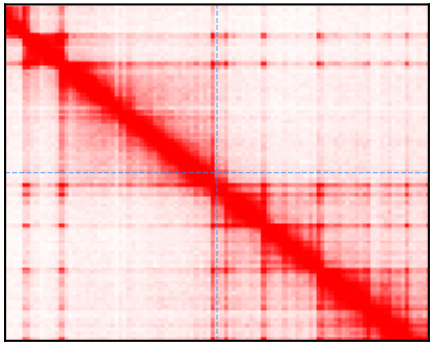
Alt (hap2) Σ=9,894



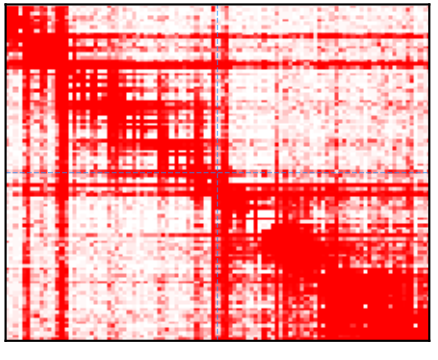
Unphased Σ=19,272

View: 82,445,000 – 82,545,000
Loop: 82,490,000 – 82,500,000

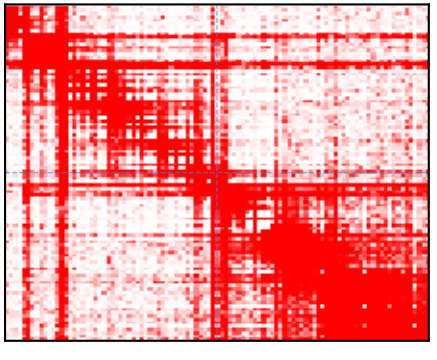
Loop 3
chr17:82,490,590-82,490,800
× chr17:82,490,000-82,500,000
dist: 4 kb
R/A: 1.04× log2FC: +0.06



Ref (hap1) Σ=3,337



Alt (hap2) Σ=3,198



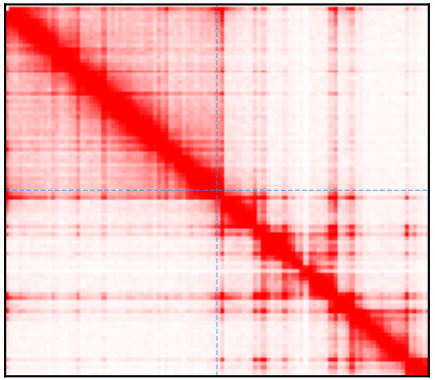
Anchor: chr19:49,443,573–49,444,053 anc_REF: 13,425 anc_ALT: 14,270

AlphaGenome pred: -0.00935 | empiric ChIP-seq: 1.35 | Hi-C: -0.09

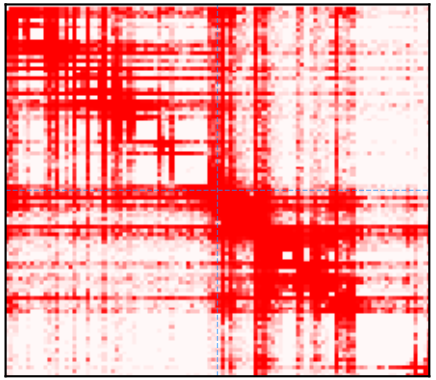
Unphased Σ=35,174

View: 49,392,500 – 49,492,500
Loop: 49,440,000 – 49,445,000

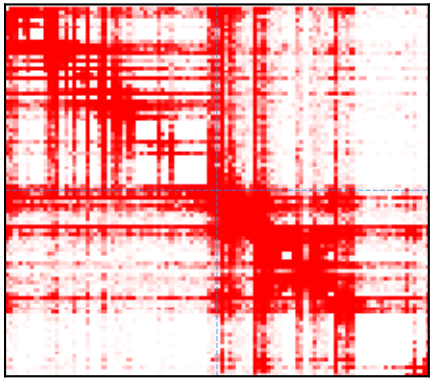
Loop 1
chr19:49,441,440-49,441,650
× chr19:49,440,000-49,445,000
dist: 1 kb
R/A: 0.94× log2FC: -0.10



Ref (hap1) Σ=3,649



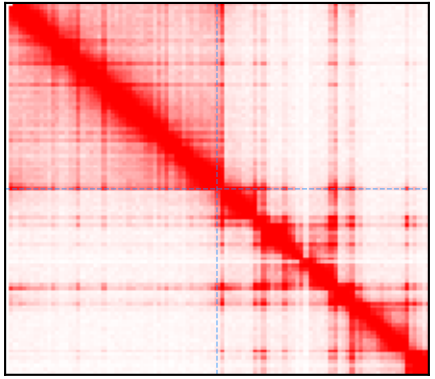
Alt (hap2) Σ=3,898



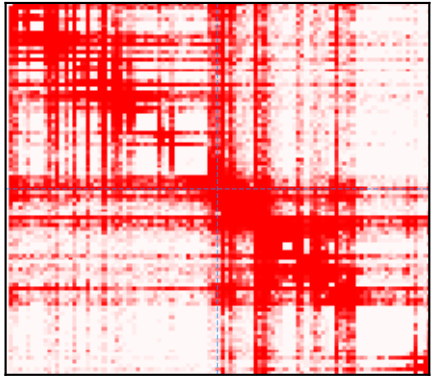
Unphased Σ=48,509

View: 49,392,500 – 49,492,500
Loop: 49,440,000 – 49,445,000

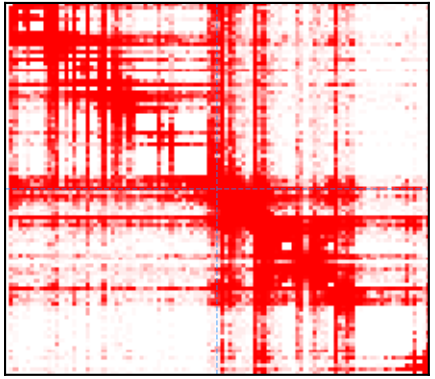
Loop 2
chr19:49,443,572-49,444,053
× chr19:49,440,000-49,445,000
dist: 1 kb
R/A: 0.94× log2FC: -0.09



Ref (hap1) Σ=9,776



Alt (hap2) Σ=10,372



Anchor: chr2:113,979,627–113,979,899 anc_REF: 4,985 anc_ALT: 6,139

AlphaGenome pred: 0.0565 | empiric ChIP-seq: -2.68 | Hi-C: -0.30

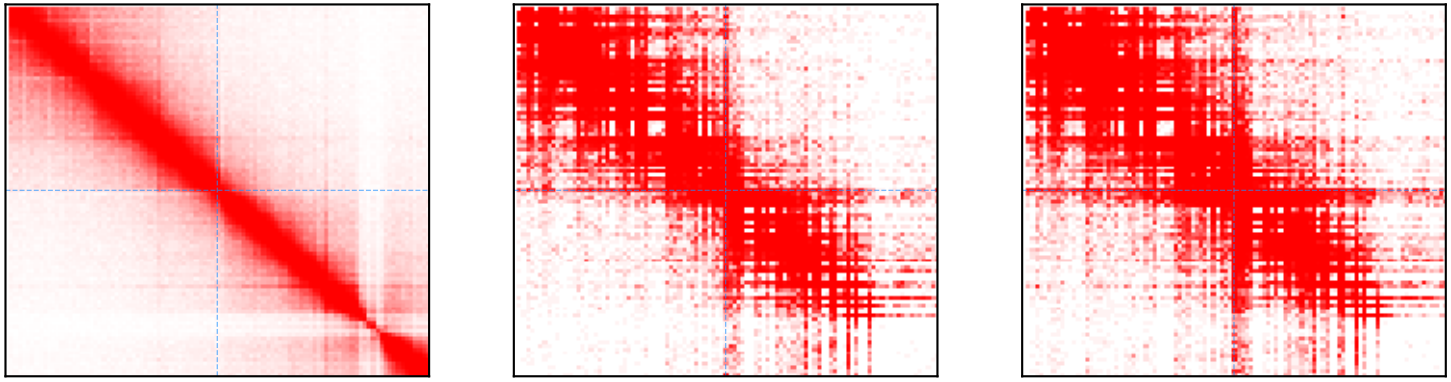
Unphased Σ=16,195

View: 113,929,500 – 114,029,500
Loop: 113,978,500 – 113,980,500

Ref (hap1) Σ=3,754

Alt (hap2) Σ=4,876

Loop 1
chr2:113,979,626-113,979,899
× chr2:113,978,500-113,980,500
dist: 0 kb
R/A: 0.77× log2FC: -0.38



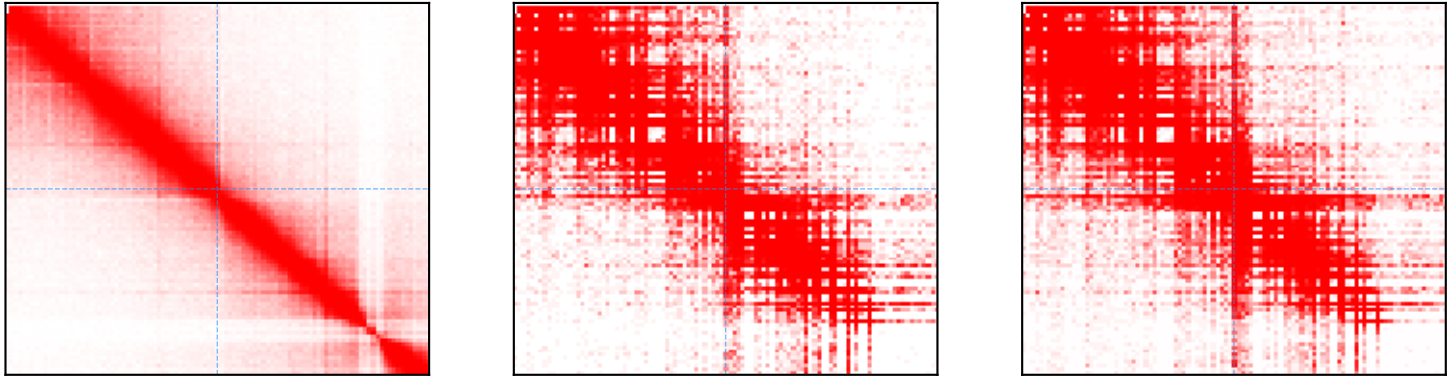
Unphased Σ=7,211

View: 113,929,500 – 114,029,500
Loop: 113,978,500 – 113,980,500

Ref (hap1) Σ=1,231

Alt (hap2) Σ=1,263

Loop 2
chr2:113,977,790-113,978,000
× chr2:113,978,500-113,980,500
dist: 2 kb
R/A: 0.97× log2FC: -0.04



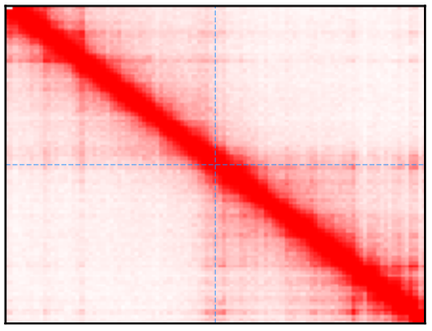
Anchor: chr3:43,996,358–43,996,504 anc_REF: 9,947 anc_ALT: 10,890

AlphaGenome pred: -0.0872 | empiric ChIP-seq: -2 | Hi-C: -0.13

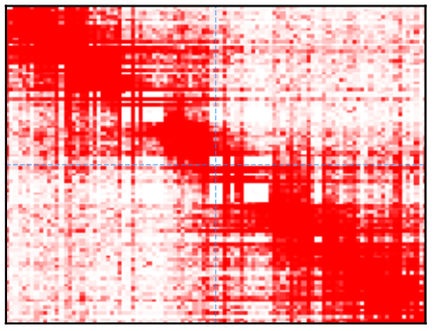
Unphased Σ=37,643

View: 43,948,000 – 44,048,000
Loop: 43,995,000 – 44,001,000

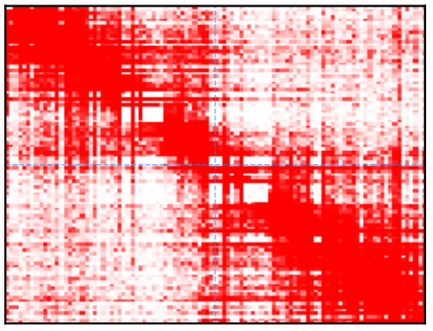
Loop 1
chr3:43,999,164-43,999,285
× chr3:43,995,000-44,001,000
dist: 1 kb
R/A: 0.87× log2FC: -0.20



Ref (hap1) Σ=570



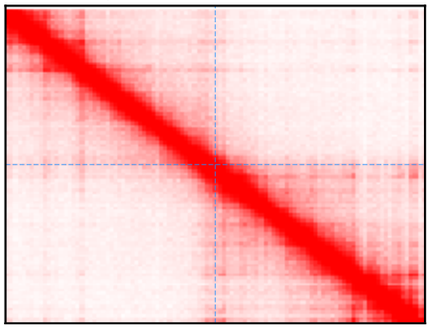
Alt (hap2) Σ=655



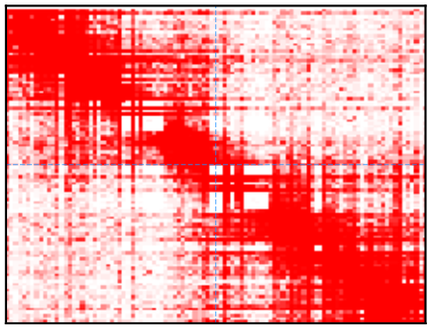
Unphased Σ=36,806

View: 43,948,000 – 44,048,000
Loop: 43,995,000 – 44,001,000

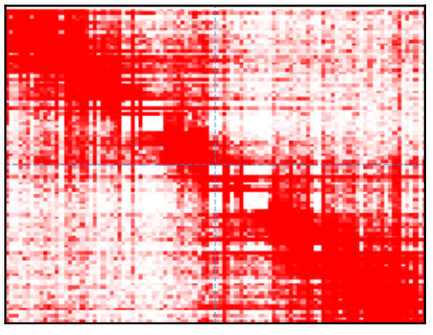
Loop 2
chr3:43,996,357-43,996,504
× chr3:43,995,000-44,001,000
dist: 2 kb
R/A: 0.91× log2FC: -0.14



Ref (hap1) Σ=4,655



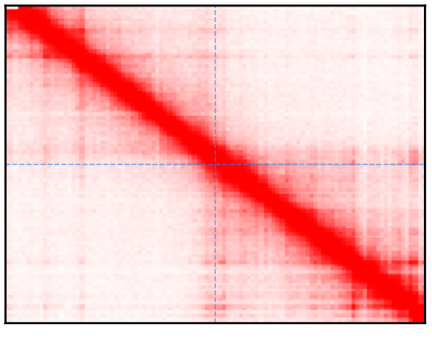
Alt (hap2) Σ=5,133



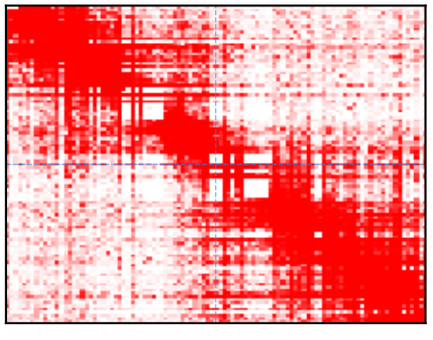
Unphased Σ=31,643

View: 43,948,000 – 44,048,000
Loop: 43,995,000 – 44,001,000

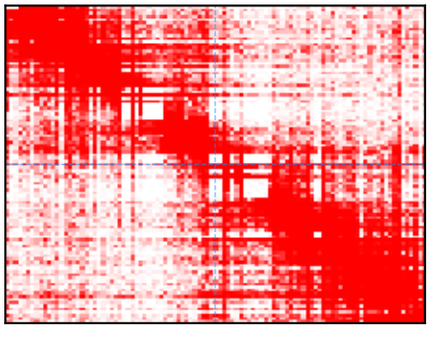
Loop 3
chr3:44,000,801-44,000,894
× chr3:43,995,000-44,001,000
dist: 3 kb
R/A: 0.93× log2FC: -0.10



Ref (hap1) Σ=1,816



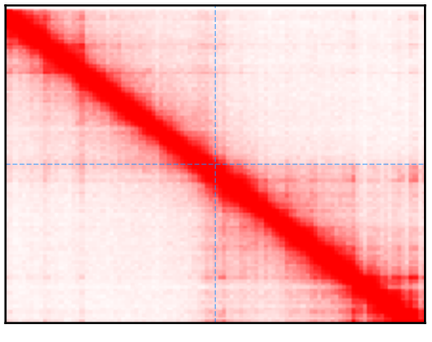
Alt (hap2) Σ=1,950



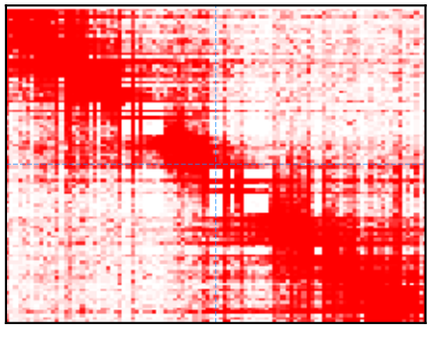
Unphased Σ=22,173

View: 43,948,000 – 44,048,000
Loop: 43,995,000 – 44,001,000

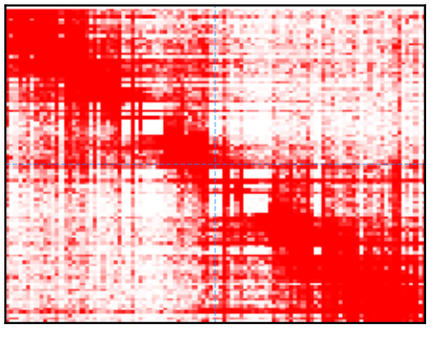
Loop 4
chr3:43,994,726-43,995,429
× chr3:43,995,000-44,001,000
dist: 3 kb
R/A: 0.92× log2FC: -0.12



Ref (hap1) Σ=2,906



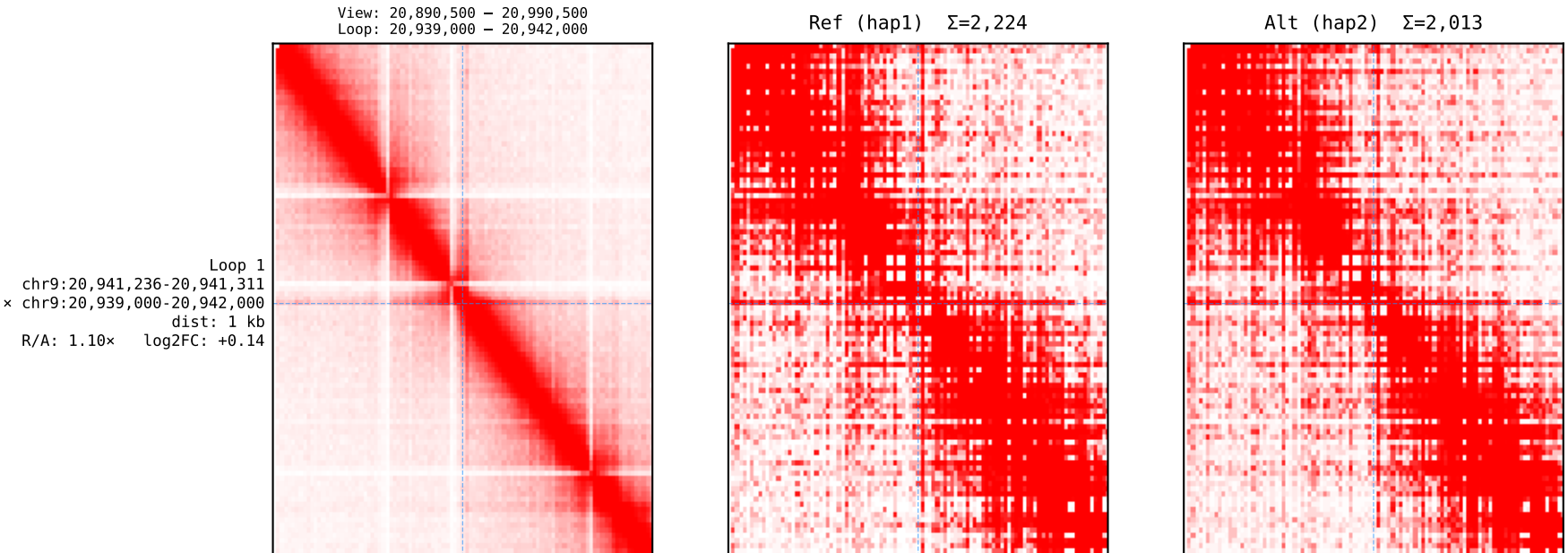
Alt (hap2) Σ=3,152



Anchor: chr9:20,941,237–20,941,311 anc_REF: 2,224 anc_ALT: 2,013

AlphaGenome pred: -0.0502 | empiric ChIP-seq: -3.25 | Hi-C: +0.14

Unphased Σ=26,666



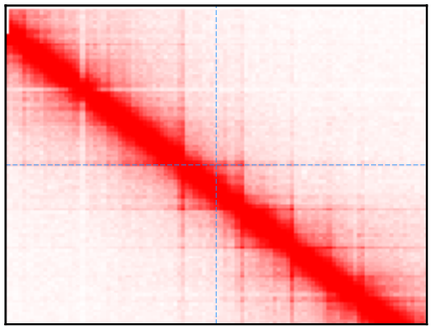
Anchor: chrX:110,145,938–110,146,092 anc_REF: 8,860 anc_ALT: 9,699

AlphaGenome pred: 0.0957 | empiric ChIP-seq: -1.25 | Hi-C: -0.13

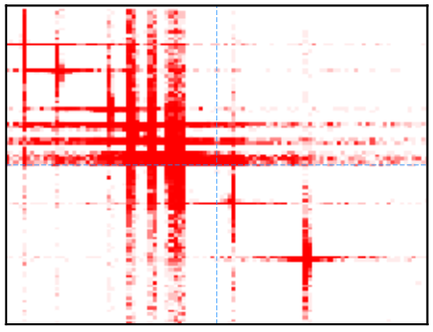
Unphased Σ=46,402

View: 110,104,500 – 110,204,500
Loop: 110,144,000 – 110,165,000

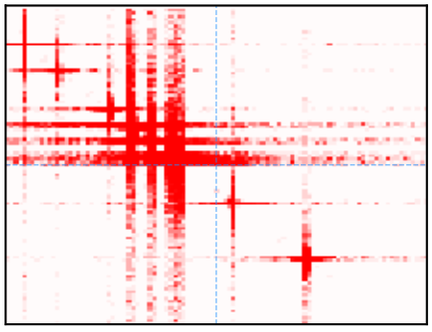
Loop 1
chrX:110,146,687-110,146,768
× chrX:110,144,000-110,165,000
dist: 8 kb
R/A: 0.91× log2FC: -0.14



Ref (hap1) Σ=4,146



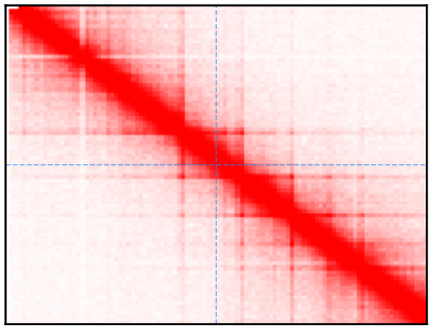
Alt (hap2) Σ=4,580



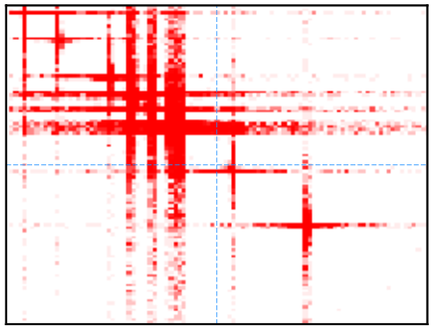
Unphased Σ=38,521

View: 110,104,500 – 110,204,500
Loop: 110,144,000 – 110,165,000

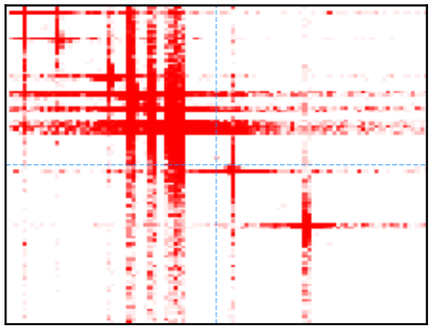
Loop 2
chrX:110,156,332-110,156,397
× chrX:110,144,000-110,165,000
dist: 2 kb
R/A: 1.18× log2FC: +0.24



Ref (hap1) Σ=247



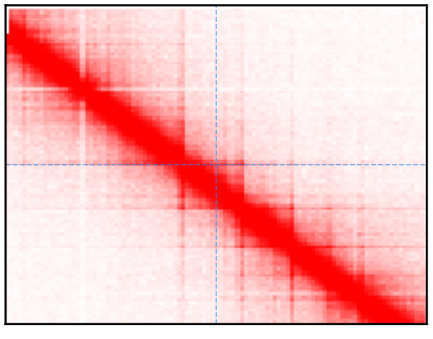
Alt (hap2) Σ=209



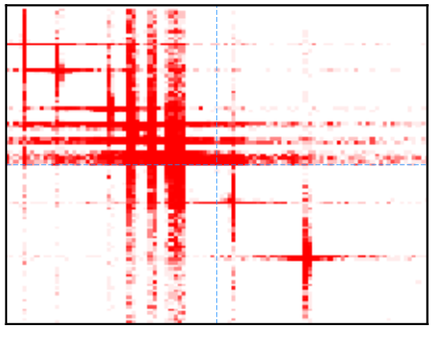
Unphased Σ=46,402

View: 110,104,500 – 110,204,500
Loop: 110,144,000 – 110,165,000

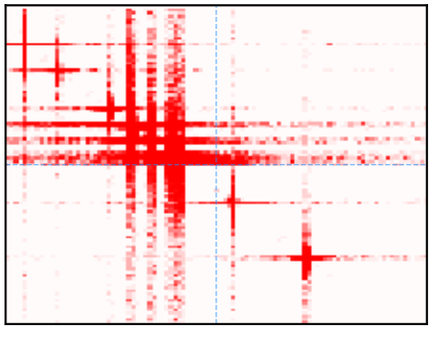
Loop 3
chrX:110,145,937-110,146,092
× chrX:110,144,000-110,165,000
dist: 8 kb
R/A: 0.91× log2FC: -0.14



Ref (hap1) Σ=4,146



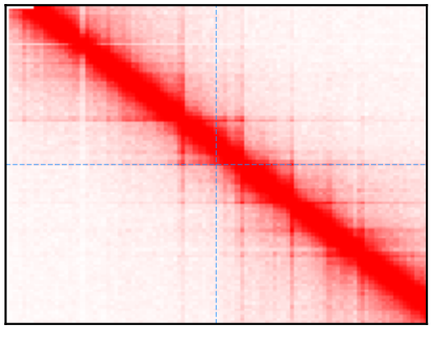
Alt (hap2) Σ=4,580



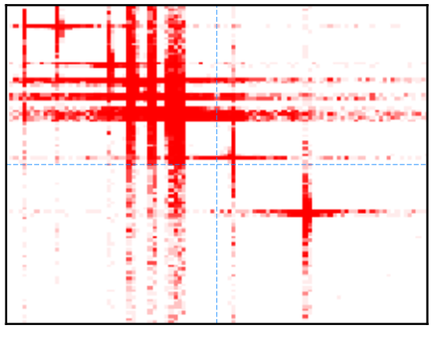
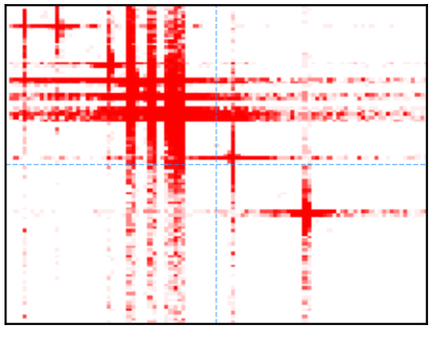
Unphased Σ=47,270

View: 110,104,500 – 110,204,500
Loop: 110,144,000 – 110,165,000

Loop 4
chrX:110,160,638-110,160,727
× chrX:110,144,000-110,165,000
dist: 6 kb
R/A: 0.97× log2FC: -0.04



Ref (hap1) Σ=321



Alt (hap2) Σ=330